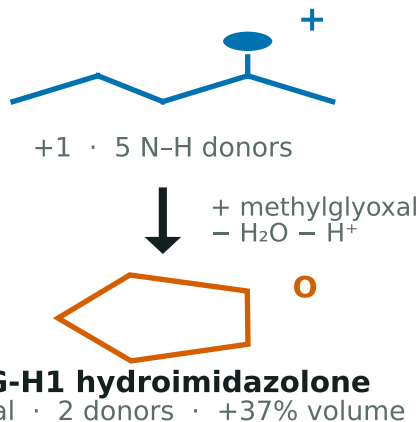


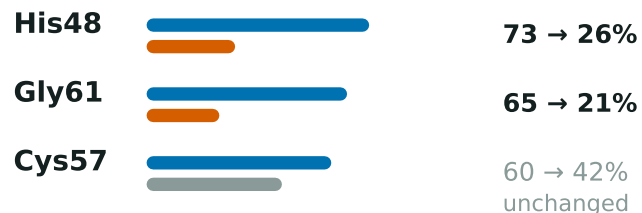
1 · THE MODIFICATION

Arg143, the catalytic arginine



2 · WHAT GLYCATION DOES

Site-143 core contact network



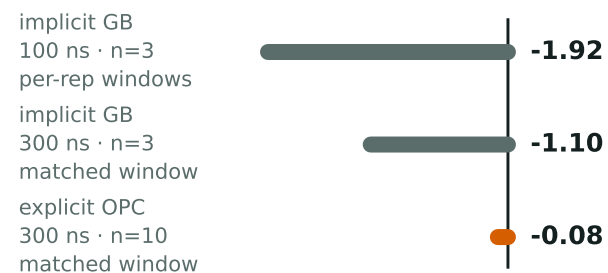
-36.1

percentage points, core composite
p = 0.0009 vs α = 0.0042
 n = 10 per group · explicit OPC water · 300 ns

3 · AND WHAT IT DEPENDS ON

Same endpoint, three methods

site H-bonds, glycated – native



The implicit-solvent result does not survive
 explicit water. Most AGE simulation work
 runs in implicit solvent.

One-residue difference

identical backbone coordinates;
 the adduct is the only variable

Replicate-level

n = 10 per group; the replicate
 is the unit, never the frame

Window-robust

significant in all five analysis
 windows, including 0–300 ns

Fully reported

negative and inconclusive
 comparisons reported alongside

Glycation at Arg143 breaks a loop-junction contact network without unfolding SOD1

Cys57, the residue implicated through its disulfide with Cys146, is the one contact that does not move.

Backbone RMSD, radius of gyration and β -sheet content are unchanged across twenty replicates: the perturbation is local, not global.